Quantization Sensitivity & Evaluation Metrics

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Accuracy is Not All You Need

- ·LLM Evaluation metrics:
  - Capability metrics: Accuracy, Perplexity
  - Distance metrics: Flips, KL-divergence

- Compression techniques should be evaluated using distance metrics
- The goal of quantized model is to mimic the behavior of the baseline model and not to become more capable

A. Dutta, S. Krishnan, N. Kwatra, and R. Ramjee, “Accuracy is Not All You Need,” Jul. 12, 2024,

Perplexity Results

Hyperparameter	LLaDA	Dream
Batch size	4	4
Monte Carlo samples	16	16
Diffusion steps / Steps	1024	1024
Generation length / Max new tokens	1024	128
Block length	1024	–
Remasking strategy	low_confidence	–
Few-shot examples	0	0

Method	Qwen3-1.7B			Qwen3-14B			Qwen3-32B			Dream-7B			LLaDA-8B		
	Mem	Wiki2 ↓	C4 ↓	Mem	Wiki2 ↓	C4 ↓	Mem	Wiki2 ↓	C4 ↓	Mem	Wiki2 ↓	C4 ↓	Mem	Wiki2 ↓	C4 ↓
Original (BF16)	3.44	16.67	19.21	29.54	8.64	12.01	65.52	7.60	10.77	15	10.53	16.36	15	9.32	16.34
SINQ (4-bit)	1.42	17.14	19.83	10.56	8.76	12.21	20.73	7.74	10.96	5.1	11.18	17.40	4.6	9.64	16.94

*C4 Results calculated only on 2048 samples

Flip Rates

SINQ 4bit quantization of Dream-7B-Base and LLaDA-8B-Base. The first/second number indicates the % of correct/incorrect answers of the baseline model that changed.

Model	ARC-Challenge	ARC-Easy	PIQA	Winogrande	MMLU	HellaSwag
LLaDA-8B-Base	2.99 / 1.45	2.27 / 1.52	1.80 / 0.87	3.95 / 3.24	6.52 / 6.65	1.20 / 0.45
Dream-7B-Base	3.67 / 2.56	2.06 / 1.35	1.85 / 1.52	3.39 / 2.29	—	—

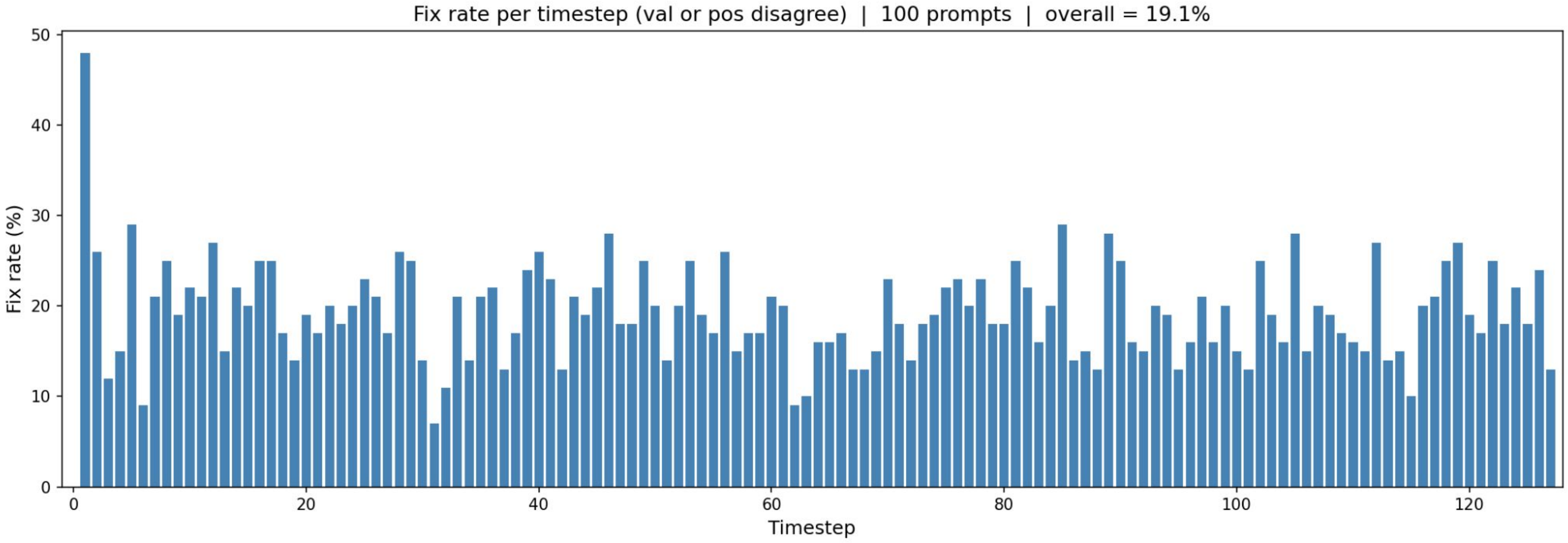
Quantization Sensitivity

- Experiment 1: Measure the fix rate, position misalignment, token value misalignment between base model and quantized model
- Experiment 2: Measure KL Divergence of the token distribution between the base and quantized model at the position to be unmasked by the unquantized model

Both experiments were conducted on 100 math prompts and 100 open ended questions.

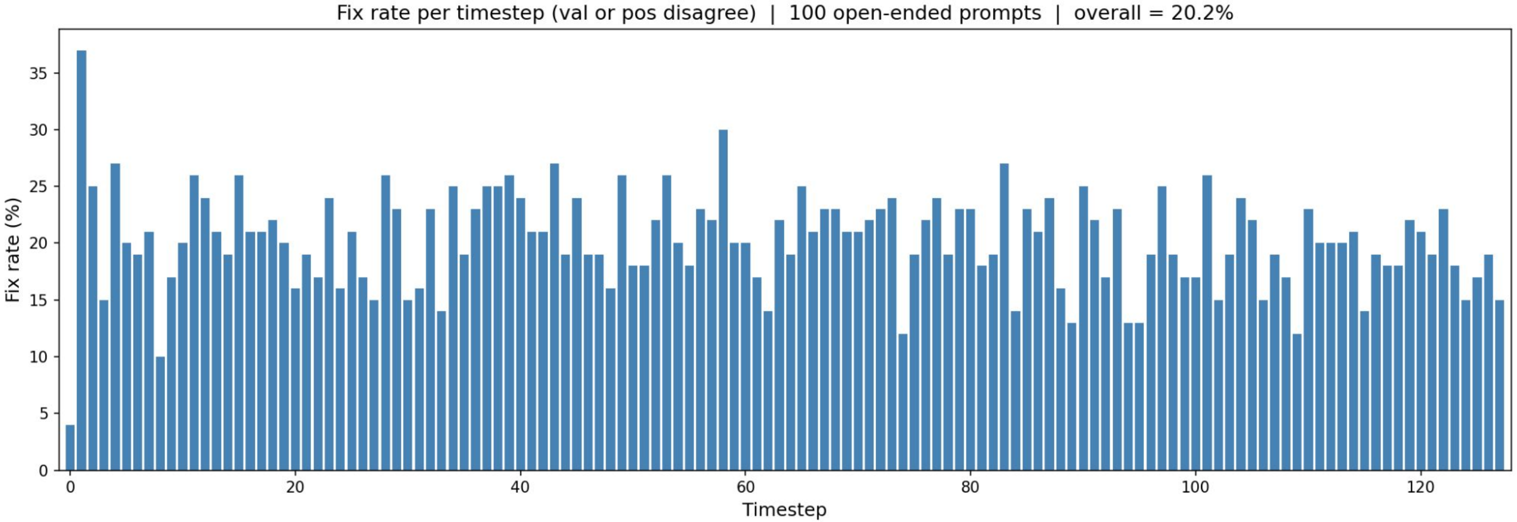
Quantization Sensitivity - Experiment 1 Results (Math)

Metric	Value
Avg position disagreements	17.52%
Avg value disagreements	2.82%
Avg fix rate (pos OR val)	19.12%

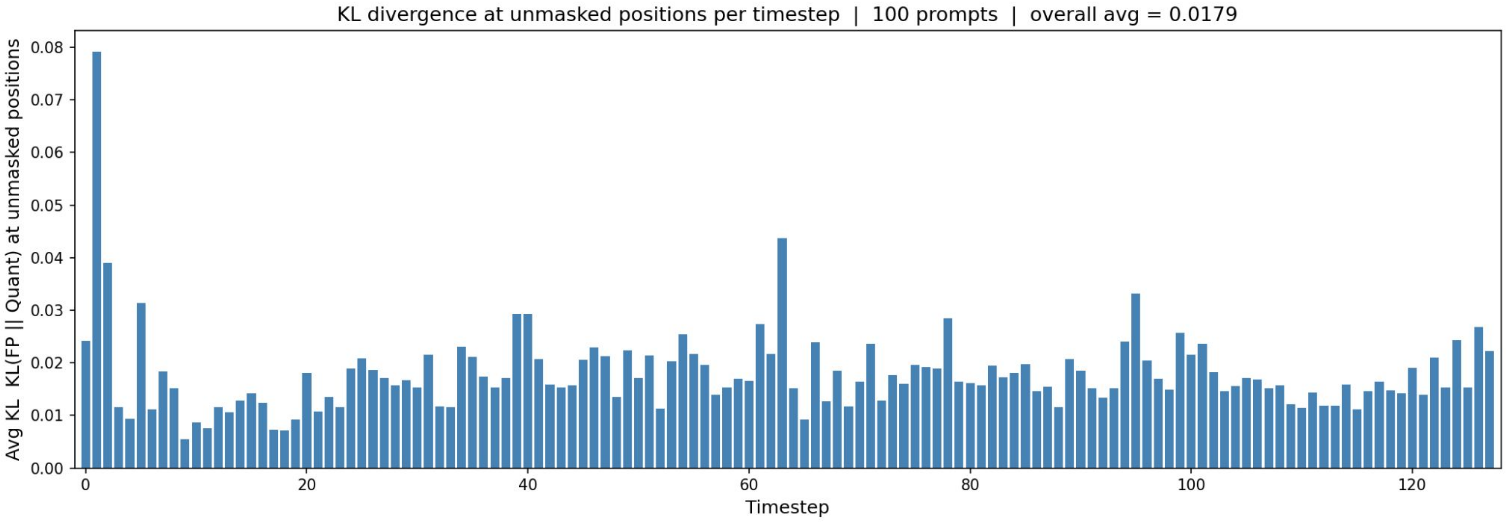


Quantization Sensitivity - Experiment 1 Results (Open Ended Questions)

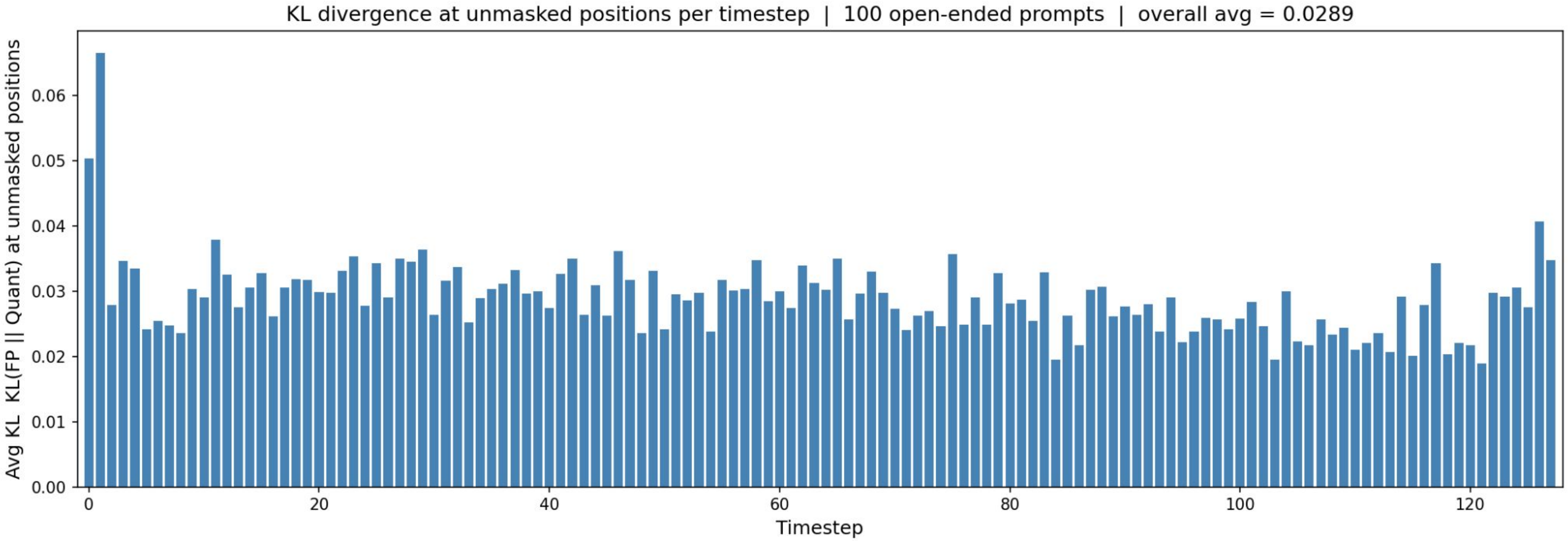
Metric	Value
Avg position disagreements	15.6%
Avg value disagreements	7.0%
Avg fix rate (pos OR val)	20.2%



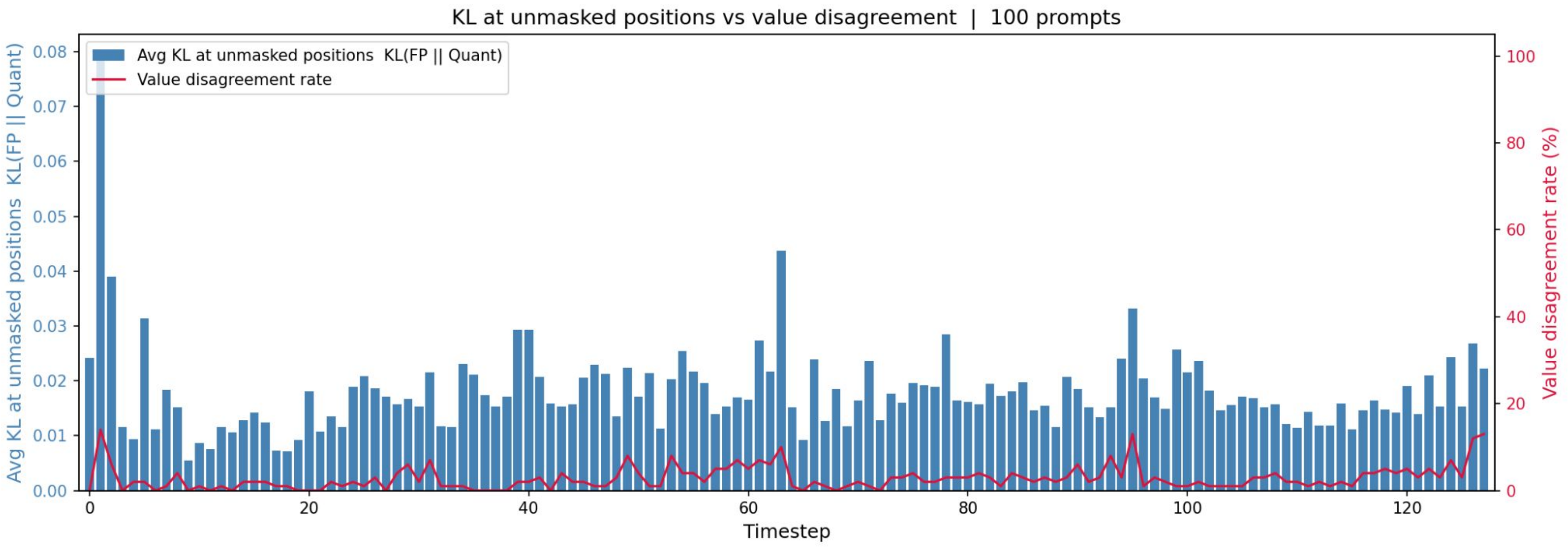
Quantization Sensitivity - Experiment 2 Results (Math)



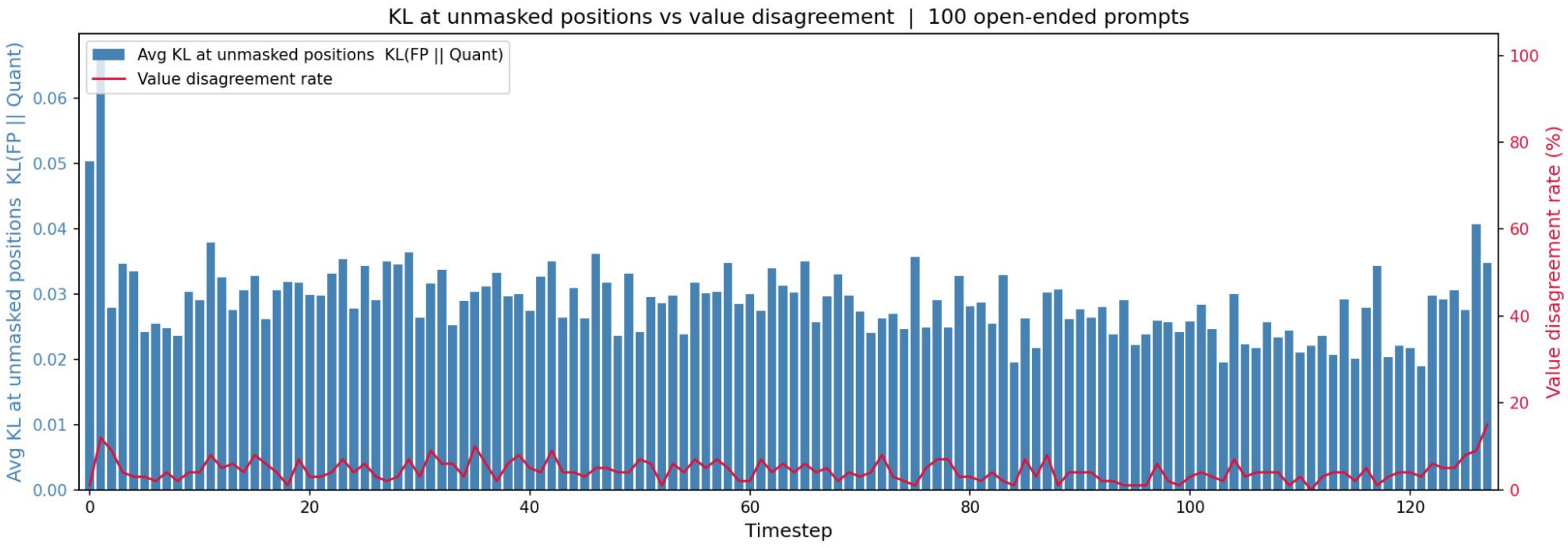
Quantization Sensitivity - Experiment 2 Results (Open Ended)



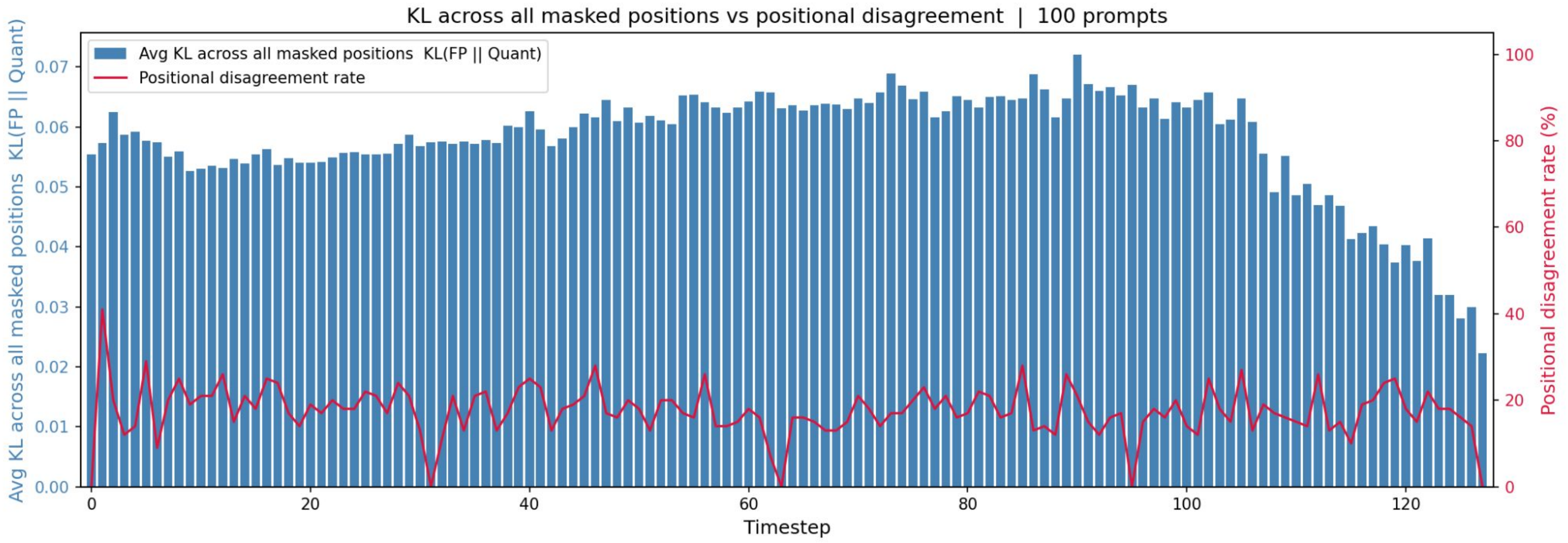
Quantization Sensitivity - KL Divergence & Value Disagreement (Math)



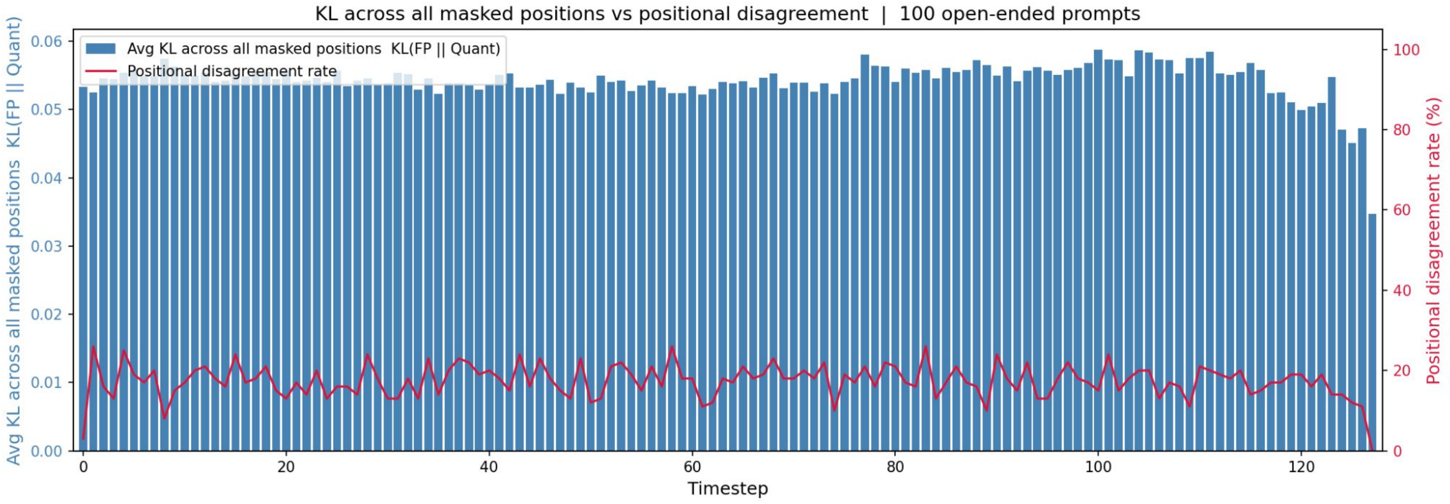
Quantization Sensitivity - KL Divergence & Value Disagreement (Open-ended)



Quantization Sensitivity - KL Divergence across positions & Positional Disagreement (Math)



Quantization Sensitivity - KL Divergence across positions & Positional Disagreement (Open-Ended)



Key Takeaways & Plans for Next Week

- Quantization error is mostly driven by positional disagreements with unquantized model
- One positional disagreement will result in having different contexts when generating the next token output which may accumulate errors
- KL Divergence is highest at earlier timesteps when the model has the least context

Plans for next week

- Analyze which layers in the model contribute mostly to these positional disagreements or confidence score perturbations
- This would help in seeing whether mixed precision approach would resolve the issue by assigning more bits to these layers of higher noise